DSC288R | GROUP #5

Predicting Poverty Susceptibility Using U.S. Census Microdata

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https://github.com/ingridalt/DSC-288R-CAPSTONE

View Video Presentation: here

Presentation Outline

01 Background & Problem Framing

02 Literature Review

03 Exploratory Data Analysis: Data and Features

04 Modeling

05 Results & Observations

06 Future Directions

BACKGROUND & PROBLEM FRAMING

Background

- Poverty is a critical public health issue
- Complexity of risk detection with vast features
- Census data is comprehensive
- Machine learning offers a scalable framework
- Early intervention, prevention, and equitable resource distribution

The "Why": **Results could inform initiatives for public programs that can assist vulnerable populations and assist those at risk to stay above the poverty line.**

Problem Framing

Beyond income: Can structural and demographic signals reliably forecast poverty risk across time?

LITERATURE

Literature Review

- In the past decades we have seen people try and solve this problem by:
  - **Supervised Classification Framing:** Poverty prediction modeled as a multi-class classification task (extreme, moderate, vulnerable, non-vulnerable) using household features education, rent, and housing quality (Venkatramolla, 2019)
  - **Predictive Feature Selection and Indicators:** Socioeconomic vulnerability is estimated through demographic & employment indicators, demonstrated by Okolie et al. (2025). Leveraging educational attainment (SCHL), employment (WKHP, ESR), health insurance coverage (HICOV), race/ethnicity (RAC1P) and disability status (DIS)
  - **Gradient Boosting Methods:** LightGBM and XGBoost enhance predictive accuracy using leaf-wise growth, regularization, and split strategies (Hu et al., 2022).

- Why ML/DL can help solve this problem:
  - Poverty outcomes depend on feature interactions (hours worked × education × employment status) that linear models cannot fully capture.

The Dataset

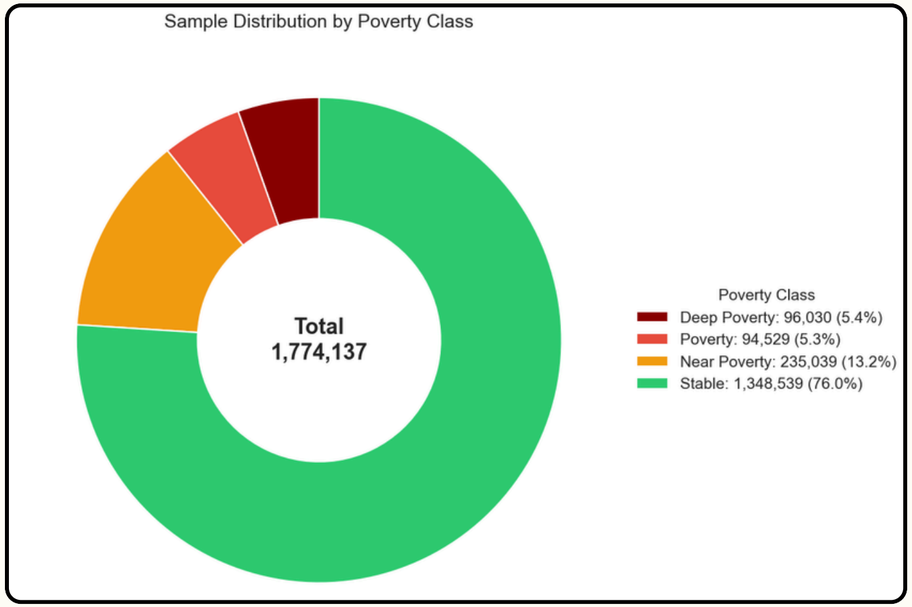
Dataset: https://www2.census.gov/programs-surveys/acs/data/pums/

- **Data Source:** Used the American Community Survey (ACS) 1-Year Public Use Microdata Sample (PUMS) for California (2018, 2019, 2021–2024) from the U.S. Census Bureau (U.S. Census Bureau, ACS PUMS).
  - Year 2020 excluded due to COVID-related anomalies.
- **Size & Structure:** ~2.3 million person-level records, 275 variables (demographic, employment, education, housing). 2024 (≈ 304,368 records (cleaned)) reserved as the test set; remaining years (≈ 1,469,769 rows records) used for training.
- **Target Variable:** Poverty ratio variable POVPIP (range 0–500). Originally categorized into 4 classes (extreme, moderate, near-poverty, non-poverty); currently exploring binary classification.
- No outliers for POVPIP were found given the nature of its distribution and range

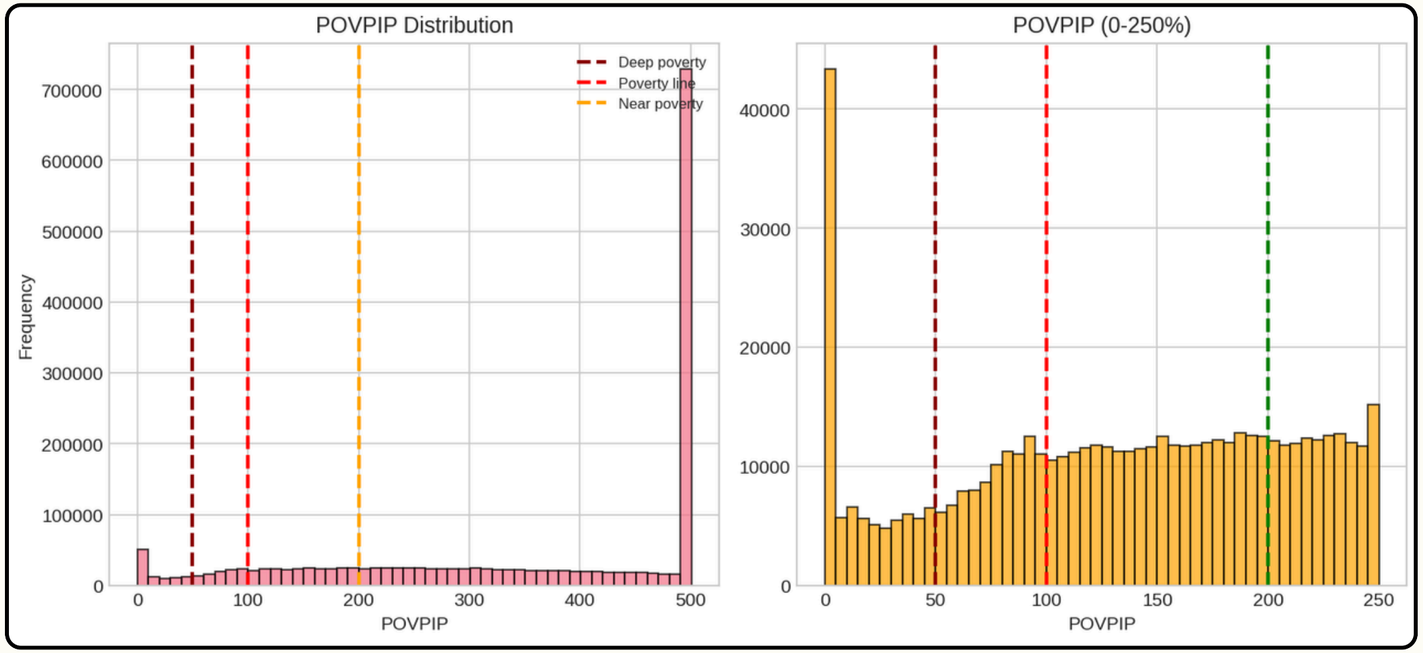

```
{
  "name": "HICOV",
  "label": "Health insurance coverage recode",
  "predicateType": "int",
  "group": "N/A",
  "limit": 0,
  "suggested-weight": "PWGTP",
  "values": {
    "item": {
      "2": "No health insurance coverage",
      "1": "With health insurance coverage"
    }
  }
```

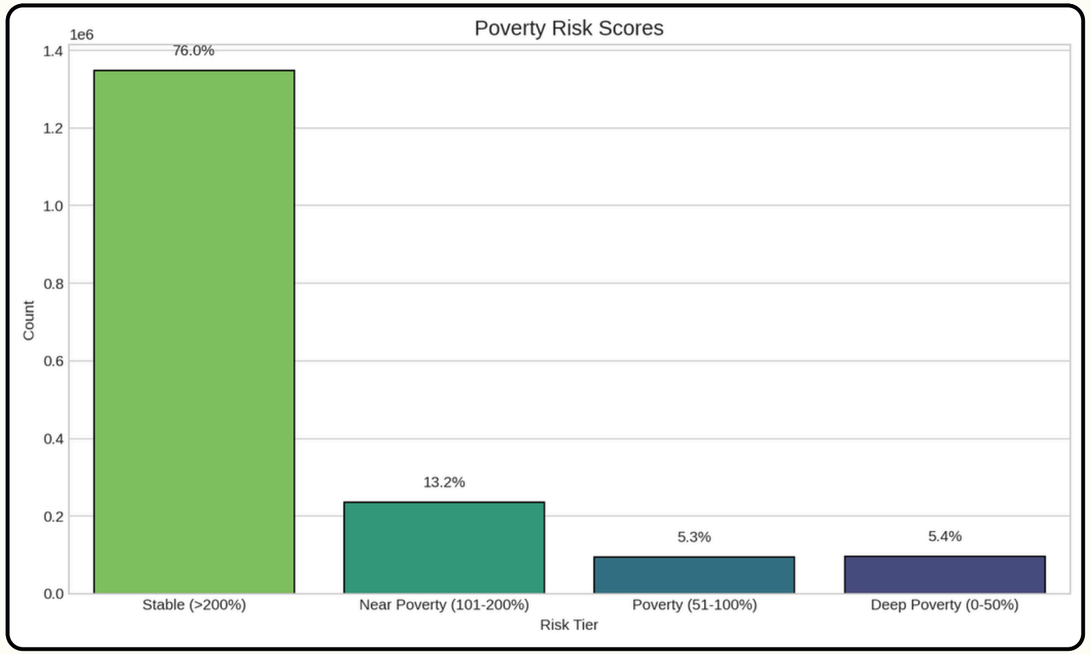
EXPLORATORY DATA ANALYSIS & FEATURES

EDA & Feature Extraction **Part I**



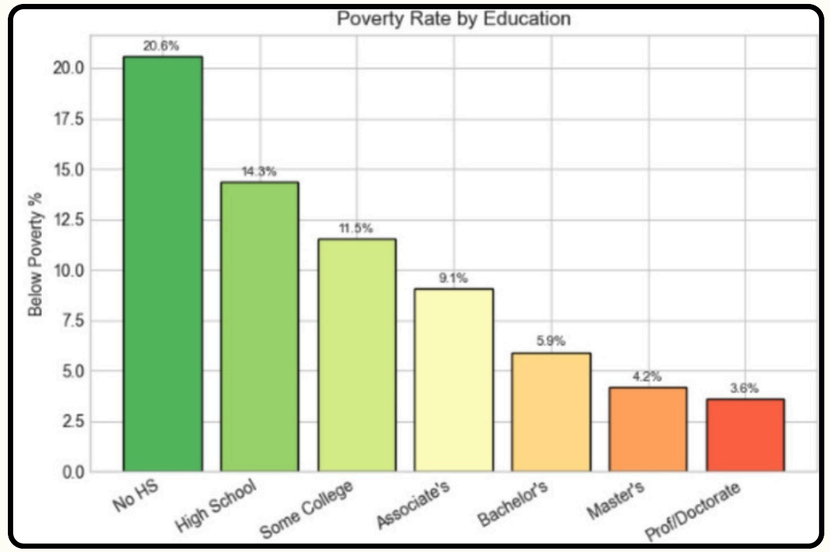
- Data Source: Census data for CA (age 19+), 2018-2024 (excluding 2020) ~1.8M records total
- Raw data contained 275 columns per year; combined into a single master dataset
- Target variable: POVPIP (income-to-poverty ratio) classified into
  - 4 poverty classes: Deep Poverty (0-50), Poverty (51-100), Near Poverty (101-200), Stable (200+)
  - OR Binary: Stable (200+) vs At Risk (0-200)
- Severe class imbalance: Stable = 76%, Near Poverty = 13.2%, Poverty = 5.3%, Deep Poverty = 5.4%
- POVPIP is on 0-500 scale, there are no outliers to handle

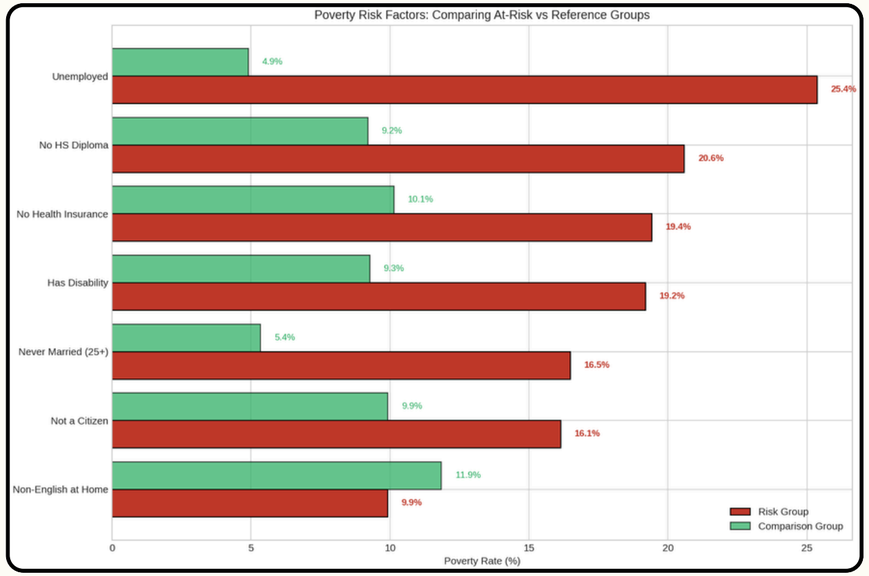


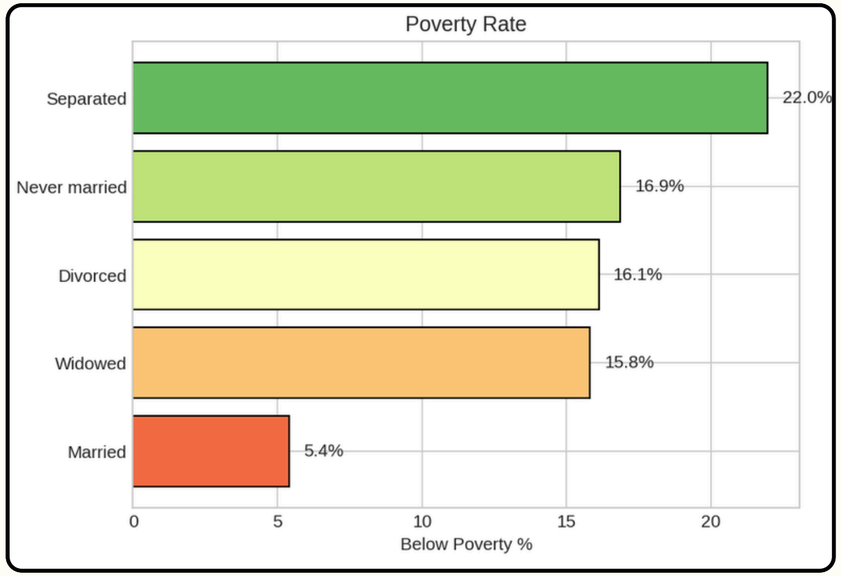


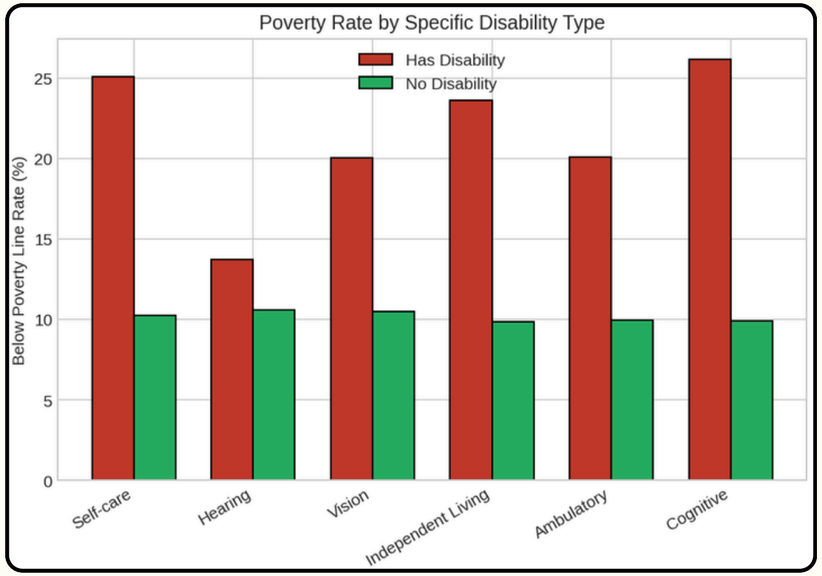
EDA & Feature Extraction **Part II**

- Dropped useless columns: Person weight replicates, allocation flags, and identifiers
- Null handling: 160 features removed with 60%+ null values
- Missing values handled with appropriate N/A imputations
- Removed 21 features with >95% single-value dominance or extremely high cardinality
- Dropped all income/benefit fields to avoid leakage; focus on non-income risk factors (demographics, education, employment, language, insurance, disability, geography)
- Correlation Analysis conducted
- EDA recommended 24 features across 8 categories
- Random undersampling used to balance classes, normalization varied



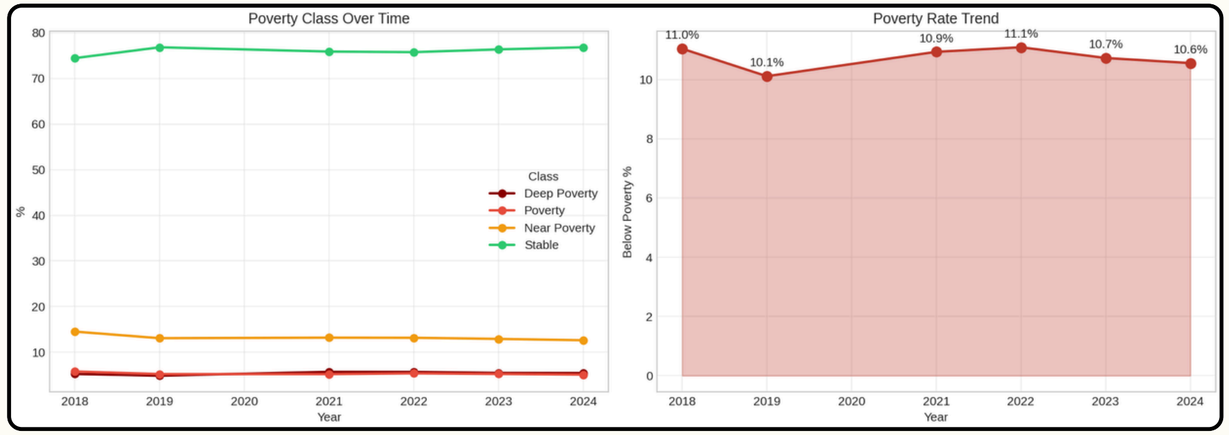


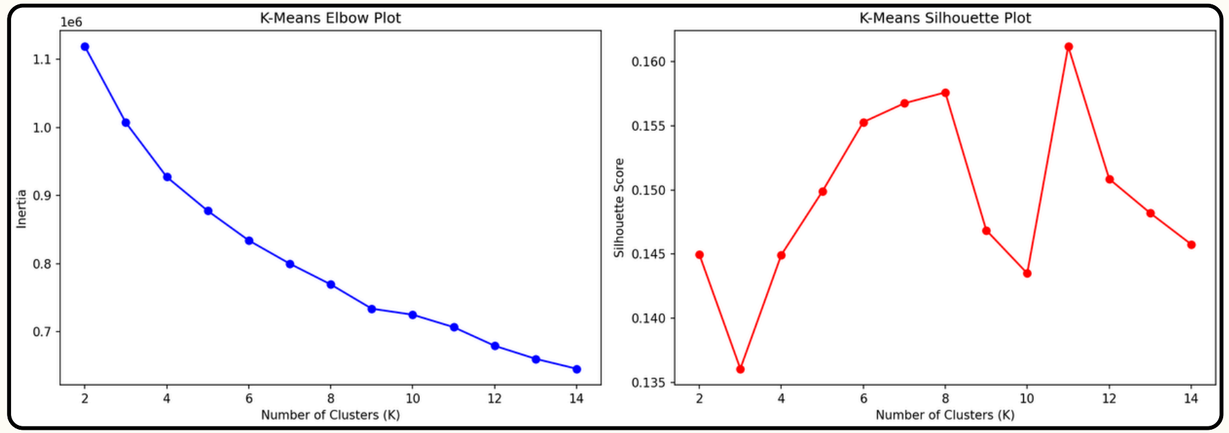


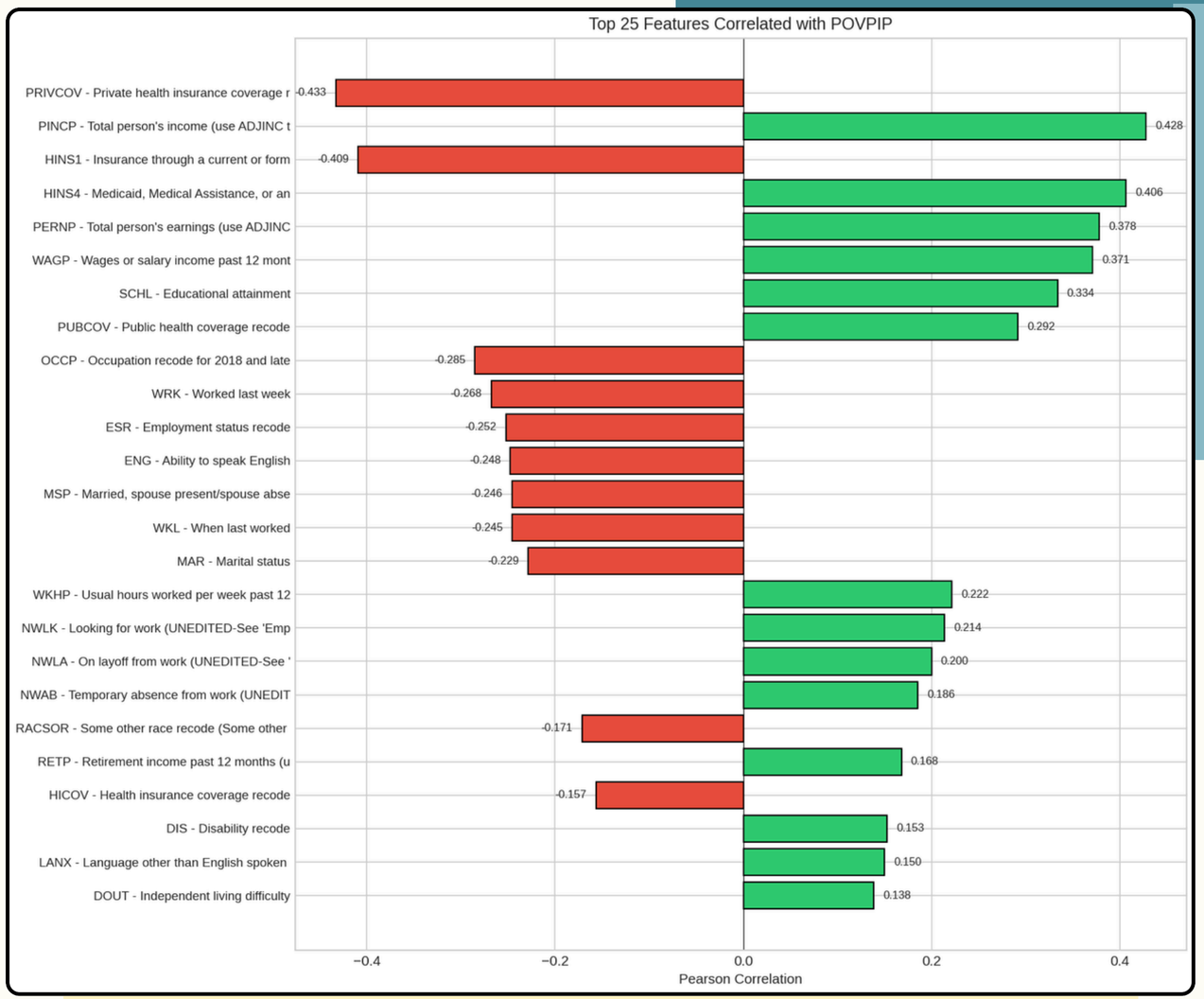


EDA & Feature Extraction **Part III**

- Train/test split is temporal: 2018-2023 train, 2024 holdout; also ran a random and saw similar performance → minimal temporal drift.
- Dimensionality Reduction and Feature Engineering: currently varies slightly from model to model, overall aiming to prune low importance features and create composite scores combining related variables to reduce dimensionality while preserving information







Original Features: 275
After Trimming Nulls: 115
After Variance Analysis: 94
After Final Correlation Analysis: 24
After Preprocessing: varies slightly from model to model

MODELS

Models

Training: 2018–2023 (Except 2020)
Testing: 2024 holdout

- **Use case framing:** Pre-screening tool for CA social services programming
- **Metric Priority:** Recall over precision for catching at-risk families

Multinomial Logistic Regression → Random Forest → XGBoost

Baseline Model

- Model used: Multiclass Logistic Regression (L2 regularized, C=0.1)
  - Standardized numeric features
  - One-hot encoded categorical variables
  - Balanced Training through downsampling used on majority class ‘Stable’
  - Using Solver ‘Saga’ (Stochastic Avg Gradient Descent) optimized for large datasets with multiple features
- Why logistic regression as our baseline?
  - Computationally inexpensive, easy to know if we have target leaking
  - Helping identify linear relationships before moving to more advanced modeling techniques
  - Interpretable through feature coefficients

```
model = LogisticRegression(
    solver='saga',
    max_iter=10000,
    penalty='l2',
    C=0.1,
    random_state=42,
    n_jobs=-1,
)
```

Random Forest

- **Preprocessing/Feature Engineering:**
  - Sparse feature matrix with engineered socioeconomic indicators like:
    - employment status, migration, nativity, housing burden, dependency structure
  - One-hot encoding + imputation
- **Nulls (Imputations):**
  - numerical (strategy=median imputation)
  - binary and categorical (strategy=mode imputation)
- **Class balancing:**
  - balanced_subsample (per-tree adaptive weighting)
  - balanced (global class weighting)
- **Validation:** Evaluation on 2024 holdout using:
  - Precision, **recall**, F1
  - Confusion matrices

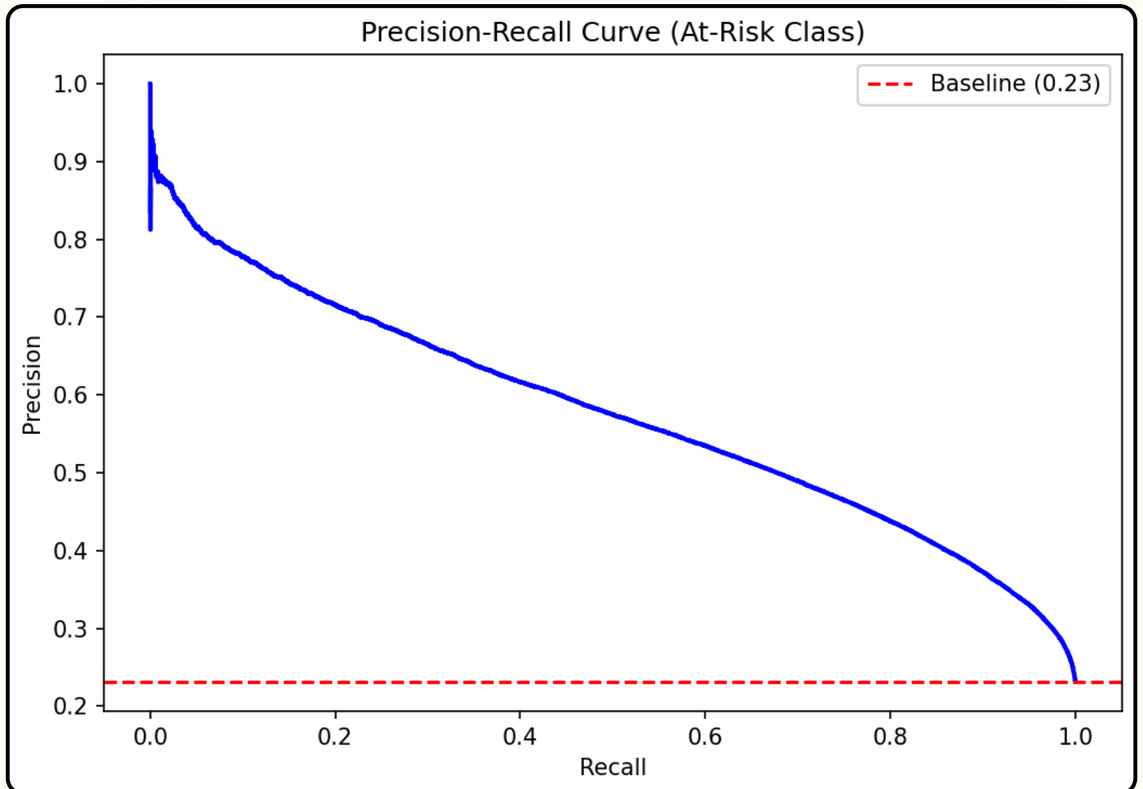
Configuration/Tuning

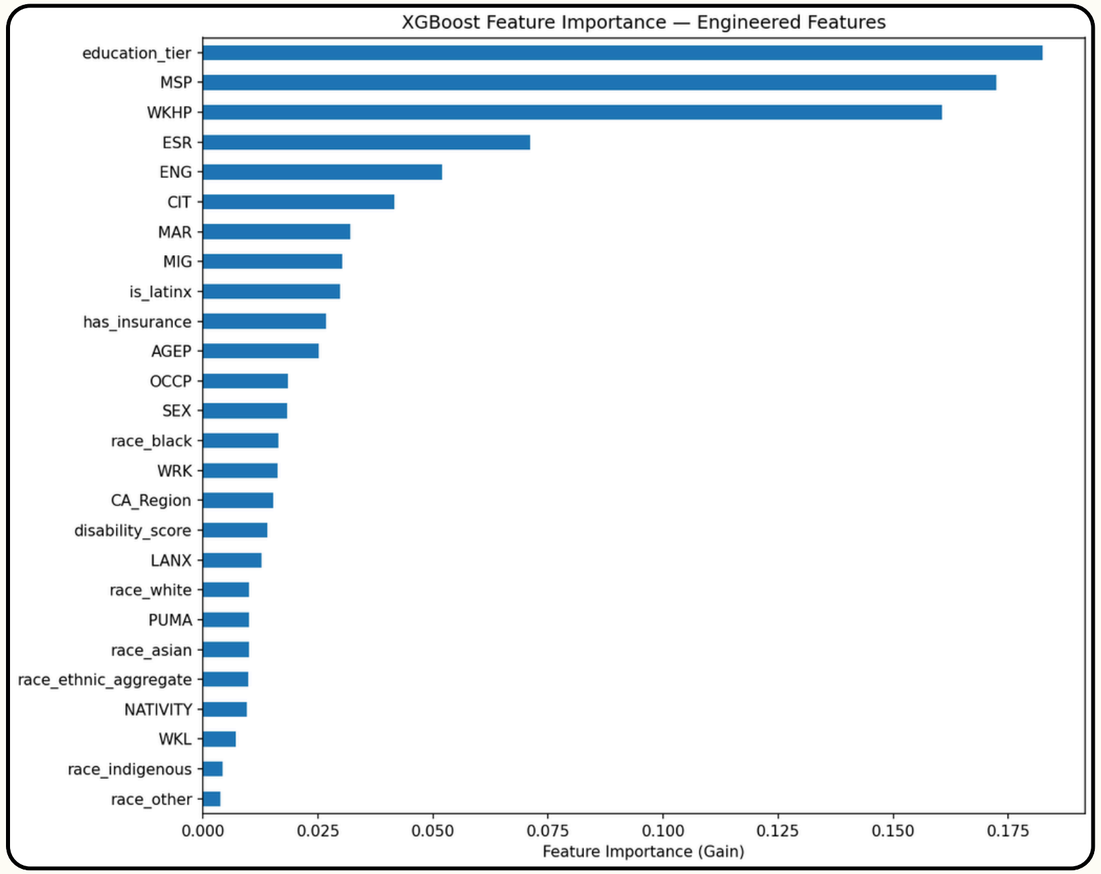
```
rf_bss = RandomForestClassifier(
    n_estimators=400,
    min_samples_split=10,
    min_samples_leaf=5,
    class_weight="balanced_subsample", # set to account for class imbalance in povpip
    random_state=42,
    n_jobs=-1
)
```

XGBoost

- **Preprocessing/Feature Engineering:** 18 features
- **Tuning:** early_stopping_rounds=30
- **Class balancing:** Random undersampling (majority class downsampled to match minority)
- **Validation:** Stratified 5-fold cross-validation with per-fold undersampling
- **Compared temporal split vs random split:** metrics within a few points; confirms robustness to year isolation.

```
Final training set (undersampled): Counter({0: 355023, 1: 355023})
```

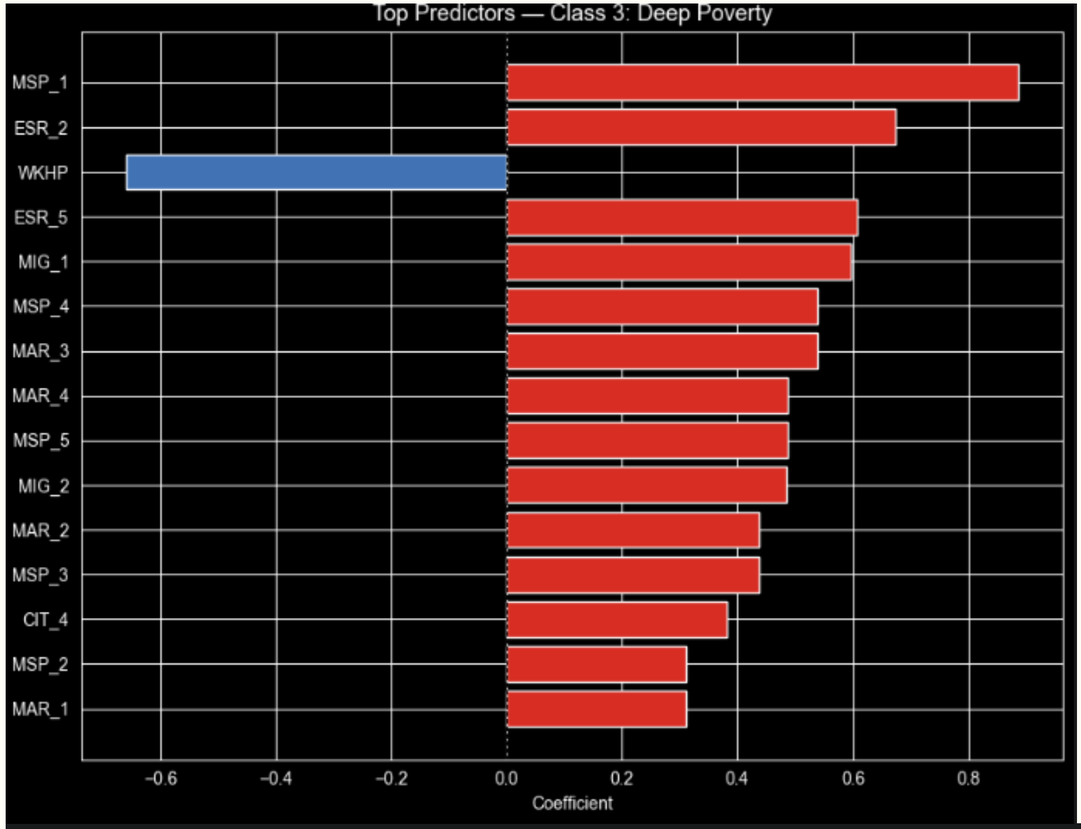




RESULTS & OBSERVATIONS

Baseline Model Results

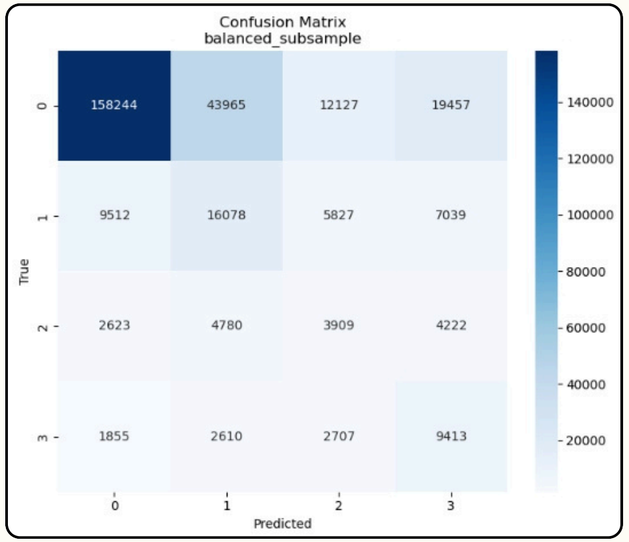
- Test Performance (2024 Set): Recall and F1-Score 0.0 for Class 2
- Key Predictors Identified: Hours worked (WKHP), education tier, marital status and employment status showed strongest coefficient influence across classes
- Implication: Linear baseline provides interpretability but lacks capacity to model nonlinear and interaction effects → motivates Random Forest and XGBoost exploration.



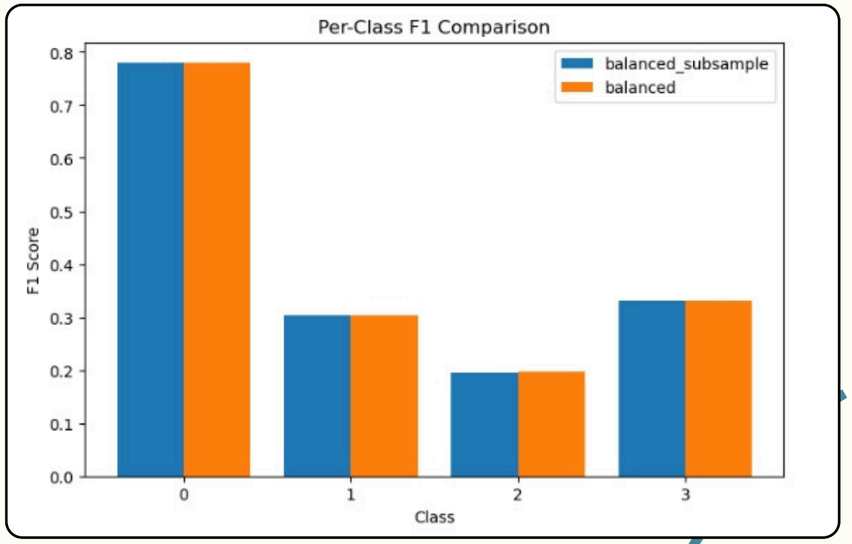
Stable (0)	0.86	0.84	0.85	233793
Near Poverty (1)	0.26	0.38	0.31	38456
Poverty (2)	0.38	0.00	0.00	15534
Deep Poverty (3)	0.30	0.34	0.32	16585
accuracy			0.71	304368
macro avg	0.45	0.39	0.37	304368
weighted avg	0.73	0.71	0.71	304368

Random Forest Results

- RF improved minority-class detection relative to the linear baseline:
  - **Macro F1 from 0.67 → 0.78**
- "balanced_subsample" produced more stable minority recall
- Improved recall for high-risk groups compared to the baseline, reducing false negatives in vulnerable populations.
- **Key Insight:** The Random Forest captures nonlinear interactions that the baseline could not, improving decision boundary flexibility.
- **Implication:** Tree-based modeling meaningfully improves minority-class learning, but residual confusion between adjacent poverty bands suggests structural overlap in predictors which motivates exploration of boosted models (XGBoost) and better feature engineering.



class	precision_subsample	precision_balanced	recall_subsample	recall_balanced	f1_subsample	f1_balanced
0	0.918773	0.918930	0.676855	0.676530	0.779475	0.779316
1	0.238429	0.238907	0.418088	0.419180	0.303676	0.304352
2	0.159096	0.160081	0.251642	0.253251	0.194943	0.196165
3	0.234557	0.233127	0.567561	0.565029	0.331935	0.330069

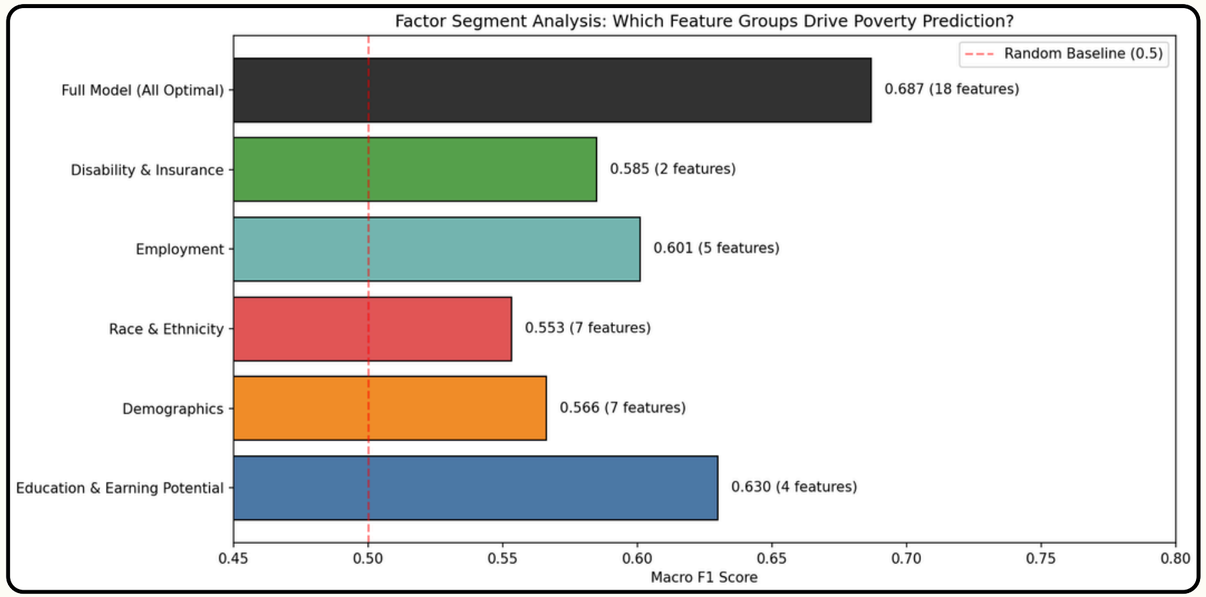


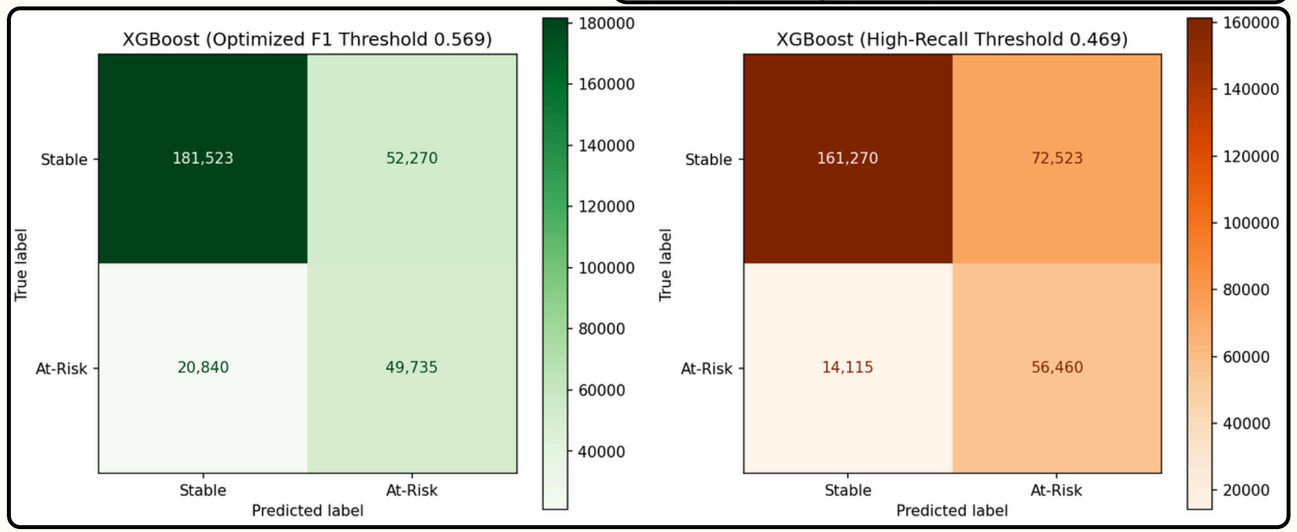
XGBoost Results

- Cross-validation: Across 5 folds - extremely stable, indicating strong generalization
- Before Feature Engineering: 67% at risk recall
- Best for use case: High-recall model (threshold 0.469) catches 4 out of 5 at-risk households (80% recall)
  - Trade-off: high # of false positives (stable households flagged for review)
- Top 3 predictors: Education tier (0.18), Marital status/spouse present (0.17), Hours worked per week (0.16) -- together account for >50% of model importance

```
=== XGBoost with Optimized Threshold (0.569) ===
              precision    recall  f1-score   support

      Stable       0.90      0.78      0.83    233793
     At-Risk       0.49      0.70      0.58     70575

    accuracy                           0.76    304368
   macro avg       0.69      0.74      0.70    304368
weighted avg       0.80      0.76      0.77    304368

Macro F1: 0.7044
Balanced Accuracy: 0.7406
```

```
=== XGBoost with High-Recall Threshold (0.469) ===
              precision    recall  f1-score   support

      Stable       0.92      0.69      0.79    233793
     At-Risk       0.44      0.80      0.57     70575

    accuracy                           0.72    304368
   macro avg       0.68      0.74      0.68    304368
weighted avg       0.81      0.72      0.74    304368

Macro F1: 0.6771
Balanced Accuracy: 0.7449
```





FUTURE DIRECTIONS

What’s Next?

Further items to complete:

- Hyperparameter tuning
- Alignment on feature engineering across models
- Continue FT Transformer Model build and Tensfor transformation

Risks & Mitigations:

- Class imbalance: at-risk population is only ~24% of data
  - Mitigation: Threshold tuning and recall-focused evaluation; undersampling shown to be effective
- Mindfulness of fairness and bias avoidance
  - Mitigation: Inclusion of underrepresented populations and referencing approaches in literature
- 60% null deletion
  - Mitigation: Exploring K-means/DBScan options
- Overall Model Performance
  - Mitigation: Training with different number of features and or feature compression/aggregates, implement grid search CV, tune hyper parameters

References

Hu, Y., Yu, Z., & Chen, X. (2022). Integrating predictive analytics with social vulnerability assessment for food insecurity research. Computers, Environment and Urban Systems, 94, 101787. https://doi.org/10.1016/j.compenvurbsys.2022.101787

Okolie, A., Lawal, O., Alumona, P., Obunadike, C., Ikhifa, M., & Akwabeng, P. (2025). Predicting food insecurity across U.S. census tracts: A machine learning analysis using the USDA Food Access Research Atlas. *International Journal of Science and Research Archive, 17*, 1156–1172. https://doi.org/10.30574/ijsra.2025.17.2.3156

Venkatramolla, Sudesh Kumar. Machine Learning and Data Science for a Household-Specific Poverty Level Prediction Task. Master's report, Kansas State University, Year. https://krex.k-state.edu/server/api/core/bitstreams/c9b10829-9856-4cb9-8333-c805146d60e8/content

CODE REVIEW

https://github.com/ingridalt/DSC-288R-CAPSTONE

Thank You!

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